Dumbbell Chart, Histograms, and Bhattacharyya Heatmap

Visualising US traffic-fatality trends and Chinese provincial COVID-19 distributions with static, interactive, and similarity-matrix plots.

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Overview

This project covers two visualisation tasks. The first reproduces a dumbbell chart of US traffic fatality rates per 100 million vehicle miles travelled across all states for 2007 and 2017, sourced from the NHTSA via an Excel workbook. The second uses 70 days of daily COVID-19 case counts for Chinese provinces to build histograms at five different bin sizes, compute pairwise Bhattacharyya coefficients, and display the resulting N-by-N similarity matrices as heatmaps.

Goals

Demonstrate how chart type and binning choices affect the information a visualisation conveys, and implement the Bhattacharyya coefficient from scratch to quantify distributional similarity between provinces.

Objectives

- Read the Dumbbell.xlsx workbook and render the dumbbell chart in both Matplotlib (static, original colours) and Plotly (interactive, hover-enabled).
- Reproduce Chongqing daily-case histograms at bin widths of 1, 5, 15, 50, and 100 cases.
- Compute the Bhattacharyya coefficient for every province pair using normalised histograms built with identical bin edges.
- Render one heatmap per bin size and analyse how the choice of bin width changes the apparent similarity structure.
- Discuss the smoothing trade-off: narrow bins exaggerate dissimilarity while wide bins wash out genuine differences.

Approach

The dumbbell chart is drawn by plotting paired scatter points per state connected by a thin grey line, with states sorted by their 2017 rate. For the histogram and heatmap work, each province's 70-day count series is binned with shared edges (0 to the dataset maximum in steps of the chosen width) and divided by total count to form a probability distribution. The Bhattacharyya coefficient is then computed as the sum of square roots of element-wise products, producing values in [0, 1]. Seaborn's heatmap function renders each N-by-N matrix with a diverging colour scale.

Outcomes

- Both Matplotlib and Plotly dumbbell charts faithfully reproduce the original Excel chart colours and state ordering.

- At bin width 1 the Bhattacharyya heatmap is nearly uniform dark, revealing almost no inter-province overlap.
- At bin widths 15 and 50 meaningful structure emerges: high-volume provinces such as Hubei remain visibly distinct while most others cluster together.
- At bin width 100 the matrix becomes uniformly bright as most counts collapse into the first one or two bins, confirming that over-smoothing erases real differences.
- The analysis shows that a moderate bin width in the range of 15 to 50 gives the most informative similarity structure for this dataset.